



Subnetting

The division of an address range of IPv4 addresses into several smaller address ranges is called **subnetting**.

A subnet is a logical segment of a network that uses IP addresses with the same network address. We can think of a subnet as a labeled entrance on a large building corridor. For example, this could be a glass door that separates various departments of a company building. With the help of subnetting, we can create a specific subnet by ourselves or find out the following outline of the respective network:

- Network address
- Broadcast address
- First host
- Last host
- Number of hosts

Let us take the following IPv4 address and subnet mask as an example:

- IPv4 Address: 192.168.12.160
- Subnet Mask: 255.255.255.192
- CIDR: 192.168.12.160/26

We already know that an IP address is divided into the **network part** and the **host part**.

Network Part

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	10 10 0000	192.168.12.160 /26
Subnet mask	1111 1111	1111 1111	1111 1111	11 00 0000	255.255.255.192
Bits	/8	/16	/24	/32	

In subnetting, we use the subnet mask as a template for the IPv4 address. From the **1**-bits in the subnet mask, we know which bits in the IPv4 address **cannot** be changed. These are **fixed** and therefore determine the "main network" in which the subnet is located.

Host Part

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	10 10 0000	192.168.12.160/26

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
Subnet mask	1111 1111	1111 1111	1111 1111	11 00 0000	255.255.255.192
Bits	/8	/16	/24	/32	

The bits in the **host part** can be changed to the **first** and **last** address. The first address is the **network address**, and the last address is the **broadcast address** for the respective subnet.

The **network address** is vital for the delivery of a data packet. If the **network address** is the same for the source and destination address, the data packet is delivered within the same subnet. If the network addresses are different, the data packet must be routed to another subnet via the **default gateway**.

The **subnet mask** determines where this separation occurs.

Separation Of Network & Host Parts

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	10 10 0000	192.168.12.160/26
Subnet mask	1111 1111	1111 1111	1111 1111	11 00 0000	255.255.255.192
Bits	/8	/16	/24	/32	

Network Address

So if we now set all bits to **0** in the **host part** of the IPv4 address, we get the respective subnet's **network address**.

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	10 00 0000	192.168.12.128 /26
Subnet mask	1111 1111	1111 1111	1111 1111	11 00 0000	255.255.255.192
Bits	/8	/16	/24	/32	

Broadcast Address

If we set all bits in the **host part** of the IPv4 address to **1**, we get the **broadcast address**.

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	10 11 1111	192.168.12.191 /26
Subnet mask	1111 1111	1111 1111	1111 1111	11 00 0000	255.255.255.192
Bits	/8	/16	/24	/32	

Since we now know that the IPv4 addresses `192.168.12.128` and `192.168.12.191` are assigned, all other IPv4 addresses are accordingly between `192.168.12.129-190`. Now we know that this subnet offers us a total of `64 - 2` (network address & broadcast address) or `62` IPv4 addresses that we can assign to our hosts.

Hosts	IPv4
Network Address	<code>192.168.12.128</code>
First Host	<code>192.168.12.129</code>
Other Hosts	<code>...</code>
Last Host	<code>192.168.12.190</code>
Broadcast Address	<code>192.168.12.191</code>

Subnetting Into Smaller Networks

Let us now assume that we, as administrators, have been given the task of dividing the subnet assigned to us into 4 additional subnets. Thus, it is essential to know that we can only divide the subnets based on the binary system.

Exponent	Value
2^0	= 1
2^1	= 2
2^2	= 4
2^3	= 8
2^4	= 16
2^5	= 32
2^6	= 64
2^7	= 128
2^8	= 256

Therefore we can divide the **64 hosts** we know by **4**. The **4** is equal to the exponent 2^2 in the binary system, so we find out the number of bits for the subnet mask by which we have to extend it. So we know the following parameters:

- Subnet: **192.168.12.128/26**
- Required Subnets: **4**

Now we increase/expand our subnet mask by **2 bits** from **/26** to **/28**, and it looks like this:

Details of	1st Octet	2nd Octet	3rd Octet	4th Octet	Decimal
IPv4	1100 0000	1010 1000	0000 1100	1000 0000	192.168.12.128 /28
Subnet mask	1111 1111	1111 1111	1111 1111	1111 0000	255.255.255.240
Bits	/8	/16	/24	/32	

Next, we can divide the **64** IPv4 addresses that are available to us into **4 parts**:

Hosts	Math	Subnets	Host range for each subnet
64	/	4	= 16

So we know how big each subnet will be. From now on, we start from the network address given to us (192.168.12.128) and add the **16** hosts **4** times:

Subnet No.	Network Address	First Host	Last Host	Broadcast Address	CIDR
1	192.168.12.128	192.168.12.129	192.168.12.142	192.168.12.143	192.168.12.12

Subnet No.	Network Address	First Host	Last Host	Broadcast Address	CIDR
2	192.168.12.144	192.168.12.145	192.168.12.158	192.168.12.159	192.168.12.14
3	192.168.12.160	192.168.12.161	192.168.12.174	192.168.12.175	192.168.12.16
4	192.168.12.176	192.168.12.177	192.168.12.190	192.168.12.191	192.168.12.17

Mental Subnetting

It may seem like there is a lot of math involved in subnetting, but each octet repeats itself, and everything is a power of two, so there doesn't have to be a lot of memorization. The first thing to do is identify what octet changes.

1st Octet	2nd Octet	3rd Octet	4th Octet
/8	/16	/24	/32

It is possible to identify what octet of the IP Address may change by remembering those four numbers. Given the Network Address: 192.168.1.1/25, it is immediately apparent that 192.168.2.4 would not be in the same network because the /25 subnet means only the fourth octet may change.

The next part identifies how big each subnet can be but by dividing eight by the network and looking at the **remainder**. This is also called **Modulo Operation (%)** and is heavily utilized in cryptology. Given our previous example of **/25**, **(25 % 8)** would be 1. This is because eight goes into 25 three times ($8 * 3 = 24$). There is a 1 leftover, which is the network bit reserved for the network mask. There is a total of eight bits in each octet of an IP Address. If one is used for the network mask, the equation becomes $2^{(8-1)}$ or 2^7 , 128. The table below contains all the numbers.

Remainder	Number	Exponential Form	Division Form
0	256	2^8	256
1	128	2^7	$256/2$
2	64	2^6	$256/2/2$
3	32	2^5	$256/2/2/2$
4	16	2^4	$256/2/2/2/2$
5	8	2^3	$256/2/2/2/2/2$
6	4	2^2	$256/2/2/2/2/2/2$
7	2	2^1	$256/2/2/2/2/2/2/2$

By remembering the powers of two up to eight, it can become an instant calculation. However, if forgotten, it may be quicker to remember to divide 256 in half the number of times of the remainder.

The tricky part of this is getting the actual IP Address range because 0 is a number and not null in networking. So in our `/25` with 128 IP Addresses, the first range is `192.168.1.0-127`. The first address is the network, and the last is the broadcast address, which means the usable IP Space would become `192.168.1.1-126`. If our IP Address fell above 128, then the `usable ip space` would be `192.168.1.129-254` (128 is the network and 255 is the broadcast).



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PRO

Question 1



Submit the decimal representation of the subnet mask from the following CIDR: 10.200.20.0/27

Submit

Question 2



Submit the broadcast address of the following CIDR: 10.200.20.0/27

Write your answer

 Submit

Question 3

 +3

 +20 ^

Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

Write your answer

 Submit

Question 4

 +3

 +20 ^

Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

Write your answer

 Submit

